

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Easy <div><div></div><div></div><div></div></div> |

Question

$x = 5$

$y = x - 8$

Which of the following points  $(x, y)$  is the solution to the given system of equations in the  $xy$ -plane?

Answer

- A.  $(0, 0)$
- B.  $(5, -3)$
- C.  $(5, -8)$
- D.  $(5, 8)$

Correct Answer: B

Rationale

Choice B is correct. A solution to a system of equations in the  $xy$ -plane is a point  $(x, y)$  that lies on the graph of each equation in the system. The first equation given is  $x = 5$ . Substituting  $5$  for  $x$  in the second given equation yields  $y = 5 - 8$ , or  $y = -3$ . It follows that in the  $xy$ -plane, the point  $(5, -3)$  lies on the graph of each equation in the system. Therefore, the solution to the given system of equations in the  $xy$ -plane is  $(5, -3)$ .

- Choice A is incorrect. The point  $(0, 0)$  doesn't lie on the graph of either equation in the given system.
- Choice C is incorrect. The point  $(5, -8)$  doesn't lie on the graph of the second equation in the given system.
- Choice D is incorrect. The point  $(5, 8)$  doesn't lie on the graph of the second equation in the given system.